

Qualification Pack



Wind Farm Engineer

QP Code: NIWE/ESS/Q1001

Version: 1.0

NSQF Level: 5

National Institute of Wind Energy || Velachery - Tambaram Main Road Pallikaranai
Chennai - 600 100 || email: pkanagavel@hotmail.com



Qualification Pack

Contents

NIWE/ESS/Q1001: Wind Farm Engineer	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
NIWE/ESS/N1001: Study on Renewable and Non- Renewable Energy Technology with Introduction to Wind Energy Scenario. Basics of Small Wind Turbine, Hybrid Systems and Offshore Wind Technology	6
NIWE/ESS/N1002: Wind Resource Assessment with practical orientation in a working wind farm	9
NIWE/ESS/N1003: Introduction to Wind Energy major components and Towers & Foundations	12
NIWE/ESS/N1004: Introduction to Aerodynamic aspects of Wind Turbines and perform practical on Hydraulics system	15
NIWE/ESS/N1005: Electrical Systems in Wind Turbines, Control and Protection system of Wind Turbines and Grid Integration Techniques	18
NIWE/ESS/N1006: Installation and Commissioning and O&M of wind farms with demonstration of Crane operations	21
NIWE/ESS/N1007: Environmental, Safety, Financial, and Policy aspects of a Wind Turbine	24
DGT/VSQ/N0102: Employability Skills (60 Hours)	27
Assessment Guidelines and Weightage	34
<i>Assessment Guidelines</i>	34
<i>Assessment Weightage</i>	35
Acronyms	37
Glossary	38



Qualification Pack

NIWE/ESS/Q1001: Wind Farm Engineer

Brief Job Description

This Qualification will enable the individual to get complete overview about all aspects of Wind Power starting from introduction to wind and its technology, wind resource assessment, installation and commissioning, operation and maintenance aspects of wind farms in a focused manner along with financial and policy aspects. They will oversee the site selection, installation, commissioning of mechanical, electrical and hydraulic components of wind turbines and take care of the safe operation for better performance. They will also guide the manpower working under them.

Personal Attributes

The individual should be able to completely overview all aspects of Wind Power. They should oversee the site selection, installation, commissioning of mechanical, electrical and hydraulic components of wind turbines and take care of the safe operation for better performance. They will also guide the manpower working under them.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [NIWE/ESS/N1001: Study on Renewable and Non- Renewable Energy Technology with Introduction to Wind Energy Scenario. Basics of Small Wind Turbine, Hybrid Systems and Offshore Wind Technology](#)
2. [NIWE/ESS/N1002: Wind Resource Assessment with practical orientation in a working wind farm](#)
3. [NIWE/ESS/N1003: Introduction to Wind Energy major components and Towers & Foundations](#)
4. [NIWE/ESS/N1004: Introduction to Aerodynamic aspects of Wind Turbines and perform practical on Hydraulics system](#)
5. [NIWE/ESS/N1005: Electrical Systems in Wind Turbines, Control and Protection system of Wind Turbines and Grid Integration Techniques](#)
6. [NIWE/ESS/N1006: Installation and Commissioning and O&M of wind farms with demonstration of Crane operations](#)
7. [NIWE/ESS/N1007: Environmental, Safety, Financial, and Policy aspects of a Wind Turbine](#)
8. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters



Qualification Pack

Sector	Environmental Science
Sub-Sector	Wind Energy
Occupation	WIND FARM ENGINEERING
Country	India
NSQF Level	5
Credits	18
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2149.0100
Minimum Educational Qualification & Experience	Pursuing 3rd year of UG (or 4th year B.E/B.Tech (Electrical/Mechanical/Electronics and Instrumentation/Civil Engineering)) OR Completed 4 year UG program (B.E/B. Tech. in Electrical/Mechanical /Electronics and Instrumentation/Civil Engineering) OR Completed 2nd year diploma after 12th (in relevant field) with 2.5 years of experience relevant experience in Electrical/Mechanical /Electronics and Instrumentation/Civil Engineering OR Completed 3 year diploma after 10th (in the relevant field) with 3 Years of experience relevant experience in Electrical/Mechanical /Electronics and Instrumentation/Civil Engineering OR Completed 3-year diploma (after 10th) (Skill Diploma in relevant field) with 2.5 years of experience relevant experience in Electrical/Mechanical /Electronics and Instrumentation/Civil Engineering OR Previous relevant Qualification of NSQF Level (4.5) with 1.5 years of experience relevant experience OR Previous relevant Qualification of NSQF Level (4) with 3 Years of experience relevant experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	22 Years
Last Reviewed On	NA



Qualification Pack

Next Review Date	31/01/2027
NSQC Approval Date	31/01/2024
Version	1.0
Reference code on NQR	QG-05-ES-01970-2024-V1-NIWE
NQR Version	1.0



Qualification Pack

NIWE/ESS/N1001: Study on Renewable and Non- Renewable Energy Technology with Introduction to Wind Energy Scenario. Basics of Small Wind Turbine, Hybrid Systems and Offshore Wind Technology

Description

Study on Renewable and Non- Renewable Energy Technology with Introduction to Wind Energy Scenario. Basics of Small Wind Turbine, Hybrid Systems and Offshore Wind Technology

Scope

The scope covers the following :

- Study on Renewable and Non- Renewable Energy Technology with Introduction to Wind Energy Scenario. Basics of Small Wind Turbine, Hybrid Systems and Offshore Wind Technology

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Develop the concepts of Renewable Energy Technology and also to know the basics of Wind Turbine



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	60	20	15	5
PC1. Develop the concepts of Renewable Energy Technology and also to know the basics of Wind Turbine	60	20	15	5
NOS Total	60	20	15	5



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIWE/ESS/N1001
NOS Name	Study on Renewable and Non- Renewable Energy Technology with Introduction to Wind Energy Scenario. Basics of Small Wind Turbine, Hybrid Systems and Offshore Wind Technology
Sector	Environmental Science
Sub-Sector	
Occupation	WIND FARM ENGINEERING, WIND FARM ENGINEERING
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024



Qualification Pack

NIWE/ESS/N1002: Wind Resource Assessment with practical orientation in a working wind farm

Description

Wind Resource Assessment with practical orientation in a working wind farm

Scope

The scope covers the following :

- Wind Resource Assessment with practical orientation in a working wind farm

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Will identify the optimal locations for maximizing energy output and improved RoI



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	60	20	15	5
PC1. Will identify the optimal locations for maximizing energy output and improved RoI	-	-	-	-
NOS Total	60	20	15	5



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIWE/ESS/N1002
NOS Name	Wind Resource Assessment with practical orientation in a working wind farm
Sector	Environmental Science
Sub-Sector	
Occupation	WIND FARM ENGINEERING
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024



Qualification Pack

NIWE/ESS/N1003: Introduction to Wind Energy major components and Towers & Foundations

Description

Introduction to Wind Energy major components and Towers & Foundations

Scope

The scope covers the following :

- Introduction to Wind Energy major components and Towers & Foundations

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Will Understand the design function and integration of wind turbine components



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	60	20	15	5
PC1. Will Understand the design function and integration of wind turbine components	60	20	15	5
NOS Total	60	20	15	5



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIWE/ESS/N1003
NOS Name	Introduction to Wind Energy major components and Towers & Foundations
Sector	Environmental Science
Sub-Sector	
Occupation	WIND FARM ENGINEERING
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024



Qualification Pack

NIWE/ESS/N1004: Introduction to Aerodynamic aspects of Wind Turbines and perform practical on Hydraulics system

Description

Introduction to Aerodynamic aspects of Wind Turbines and perform practical on Hydraulics system

Scope

The scope covers the following :

- Introduction to Aerodynamic aspects of Wind Turbines and perform practical on Hydraulics system

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Encompass a deep understanding of the principles, design consideration and technological advances related to wind turbine blade.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	60	20	15	5
PC1. Encompass a deep understanding of the principles, design consideration and technological advances related to wind turbine blade.	-	-	-	-
NOS Total	60	20	15	5



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIWE/ESS/N1004
NOS Name	Introduction to Aerodynamic aspects of Wind Turbines and perform practical on Hydraulics system
Sector	Environmental Science
Sub-Sector	
Occupation	WIND FARM ENGINEERING
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024



Qualification Pack

NIWE/ESS/N1005: Electrical Systems in Wind Turbines, Control and Protection system of Wind Turbines and Grid Integration Techniques

Description

Electrical Systems in Wind Turbines, Control and Protection system of Wind Turbines and Grid Integration Techniques

Scope

The scope covers the following :

- Electrical Systems in Wind Turbines, Control and Protection system of Wind Turbines and Grid Integration Techniques

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. To study the electrical systems in wind turbine and also to cover a wide range of technical, economic and practical aspects of grid integration techniques.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	60	20	15	5
PC1. To study the electrical systems in wind turbine and also to cover a wide range of technical, economic and practical aspects of grid integration techniques.	-	-	-	-
NOS Total	60	20	15	5



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIWE/ESS/N1005
NOS Name	Electrical Systems in Wind Turbines, Control and Protection system of Wind Turbines and Grid Integration Techniques
Sector	Environmental Science
Sub-Sector	
Occupation	WIND FARM ENGINEERING
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024



Qualification Pack

NIWE/ESS/N1006: Installation and Commissioning and O&M of wind farms with demonstration of Crane operations

Description

Installation and Commissioning and O&M of wind farms with demonstration of Crane operations

Scope

The scope covers the following :

- Installation and Commissioning and O&M of wind farms with demonstration of Crane operations

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Study about the project planning and site preparation, in addition to learning about wind turbine transportation and logistics



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	60	20	15	5
PC1. Study about the project planning and site preparation, in addition to learning about wind turbine transportation and logistics	-	-	-	-
NOS Total	60	20	15	5



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIWE/ESS/N1006
NOS Name	Installation and Commissioning and O&M of wind farms with demonstration of Crane operations
Sector	Environmental Science
Sub-Sector	
Occupation	WIND FARM ENGINEERING, WIND FARM ENGINEERING
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024



Qualification Pack

NIWE/ESS/N1007: Environmental, Safety, Financial, and Policy aspects of a Wind Turbine

Description

Environmental, Safety, Financial, and Policy aspects of a Wind Turbine.

Scope

The scope covers the following :

- Environmental, Safety, Financial, and Policy aspects of a Wind Turbine.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Will understand the impacts, regulations, and best practices including environmental aspects associated with wind energy development



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	60	20	15	5
PC1. Will understand the impacts, regulations, and best practices including environmental aspects associated with wind energy development	-	-	-	-
NOS Total	60	20	15	5



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIWE/ESS/N1007
NOS Name	Environmental, Safety, Financial, and Policy aspects of a Wind Turbine
Sector	Environmental Science
Sub-Sector	
Occupation	WIND FARM ENGINEERING, WIND FARM ENGINEERING
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024



Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:



Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.



Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

- The assessment strategy includes the processes involved in identifying, gathering and interpreting information to evaluate the learner on the required competencies of the program.
 - The assessment for the question paper will be based on the question bank created for the purpose.
 - The assessment agency (NIWE) will create separate set of questions for practical exam for every student at each examination/training center.
- In case of unsuccessful completion, the trainee may seek reassessment on the qualification.
- Every trainee should score a minimum of 65% of marks to successfully clear the assessment.
 - The Assessment agency will deploy the certified Assessor for executing the assessment.
 - Physical verification of the centre to be assessed.
 - Checking the allotted time to the candidates to complete Theory & Practical Assessment is correct.



Qualification Pack

- ☐ Checking the type of assessment whether offline or online.
- ☐ Checking the availability of lab equipment for a particular job role.
- ☐ Verification of question papers that was prepared by the subject matter experts.
- ☐ Verifying whether the question papers are prepared as per the NSQF Level 5.
- ☐ Reporting of Time-stamp & geotag of the assessor from the assessment location.
- ☐ Biometric or manual attendance sheet (duly signed by TP) of the trainees during the training period.
- ☐ Time-stamped & geotagged assessment (Theory + Practical + Viva) photographs & videos.
- ☐ Surprise visit to the assessment location.

Minimum Aggregate Passing % at QP Level : 65

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIWE/ESS/N1001.Study on Renewable and Non- Renewable Energy Technology with Introduction to Wind Energy Scenario. Basics of Small Wind Turbine, Hybrid Systems and Offshore Wind Technology	60	20	15	5	100	15
NIWE/ESS/N1002.Wind Resource Assessment with practical orientation in a working wind farm	60	20	15	5	100	15
NIWE/ESS/N1003.Introduction to Wind Energy major components and Towers & Foundations	60	20	15	5	100	15



Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIWE/ESS/N1004.Introduction to Aerodynamic aspects of Wind Turbines and perform practical on Hydraulics system	60	20	15	5	100	15
NIWE/ESS/N1005.Electrical Systems in Wind Turbines, Control and Protection system of Wind Turbines and Grid Integration Techniques	60	20	15	5	100	10
NIWE/ESS/N1006.Installation and Commissioning and O&M of wind farms with demonstration of Crane operations	60	20	15	5	100	10
NIWE/ESS/N1007.Environmental, Safety, Financial, and Policy aspects of a Wind Turbine	60	20	15	5	100	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	440	170	105	35	750	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.